

# VASILEIOS KARAGIANNIDIS

Junior Software / Embedded Engineer · Robotics Researcher

Thessaloniki, Greece • +30 694 053 4726 • [billykaragian1502@gmail.com](mailto:billykaragian1502@gmail.com) • [github.com/karagiancode](https://github.com/karagiancode)

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## PROFESSIONAL SUMMARY

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Undergraduate Computer Engineering student with hands-on experience in Python, PyTorch, embedded systems (ESP32, Raspberry Pi) and active research work in robotics and deep learning. My bachelor thesis involves the development of a DIY robotic arm built around a Raspberry Pi 5 + 2× Pico and multiple stepper motors, while I'm also working on computer vision and mobile manipulation with the Boston Dynamics Spot robot on NVIDIA Isaac Sim. I have additionally completed an academic research project comparing multimodal LLMs with fine-tuned Vision Transformers for medical imaging. Vice President of the IEEE IHU Student Branch and co-organizer of the IEEEetcon 2026 conference.

**Looking for:** Internship or Junior position in Software Development, Embedded Systems, Robotics or IoT teams — full-time, on-site/hybrid in Thessaloniki or remote.

## TECHNICAL SKILLS

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**Programming Languages:** Python, Java, JavaScript, SQL, C/C++ (embedded)

**Frameworks & Libraries:** JavaFX, PyTorch, HuggingFace Transformers, CustomTkinter, Tkinter, scikit-learn, pandas, PIL

**Databases & Cloud:** PostgreSQL, JDBC, Firebase (real-time DB & cloud sync)

**Machine Learning:** Vision Transformers (ViT), Multimodal LLMs (LLaVA), fine-tuning, evaluation metrics, ISIC 2017

**Embedded & Hardware:** Raspberry Pi 5, Raspberry Pi Pico, ESP32, stepper motors, circuit design & analysis

**Robotics & Simulation:** NVIDIA Isaac Sim, Boston Dynamics Spot SDK, Forward/Inverse Kinematics, Computer Vision, Mobile Manipulation

**Tools & Platforms:** Git, GitHub, Linux, VS Code, PhotoRec, file system automation

**Areas of Interest:** Robotics, Deep Learning, IoT, Embedded Systems, Computer Vision, Automation

## BACHELOR THESIS

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**DIY Robotic Arm** — International Hellenic University 2025 – 2026 (in progress)

Design and development of a DIY robotic arm capable of stably lifting objects from 500g up to 2kg, with the sensitivity required to grip an egg without breaking it.

- Working on the arm's kinematics — searching for the optimal approach (Forward / Inverse Kinematics)
- Hardware: multiple stepper motors controlled by a Raspberry Pi 5 and 2× Raspberry Pi Pico
- Developing control software and coordinating communication between the controllers
- Goal: stability + sensitivity across a wide range of object weights and fragility

## PROFESSIONAL & ACADEMIC EXPERIENCE

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**Researcher — Robotic Simulation & Vision** — International Hellenic University 2025 – Present

- Research project on the Boston Dynamics Spot quadruped robot in NVIDIA Isaac Sim simulation
- Working on the Spot vision system and implementing mobile manipulation tasks — autonomous navigation and object pick & place via the arm
- Continuous deepening in configuration, simulation workflow and robotic application development

**Vice President** — IEEE IHU Student Branch 2025 – 2026

- Co-organizer of the two-day IEEEetcon 2026 conference — inviting and coordinating speakers, liaising with companies and academic teams, securing sponsorships
- Co-organized the branch's participation in the international IEEE Extreme competition (October 2025) — secured sponsorships and achieved a high final ranking

- Designed, organized and delivered technical workshops for students: 'Introduction to Microcontrollers' and 'Microcontrollers & Robotics'

**Kitchen Coach** — KFC Thessaloniki

2020 – 2025

- Training and mentoring of new kitchen staff — onboarding & coaching
- Kitchen team supervisor and shift workflow management
- Mediator for store-level operational issue resolution
- Enforcement of Food Safety standards

## SELECTED PROJECTS

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**Multimodal Medical Image Classification** — Academic research project: comparison of zero-shot multimodal LLM (LLaVA) against a fine-tuned Vision Transformer (ViT) on dermatological image classification (ISIC 2017). The fine-tuned ViT achieved 78% accuracy — significantly higher than the zero-shot LLaVA (39%). Documented in an IEEE-format paper.

*Tech:* Python, PyTorch, HuggingFace Transformers, ViT, LLaVA, ISIC 2017

**Spot — Vision & Mobile Manipulation** — Research project on the Boston Dynamics Spot quadruped: vision system processing, autonomous navigation and object pick & place via the arm, in Isaac Sim simulation.

*Tech:* NVIDIA Isaac Sim, Boston Dynamics Spot SDK, Python, Computer Vision

**Fast-Food-App** — Full-stack desktop order management application for a fast-food store with persistent storage — inspired by my KFC experience.

*Tech:* Java, JavaFX, PostgreSQL, JDBC

**Security-System-with-ESP32** — ESP32-based security system — sensors, real-time alerts and control.

*Tech:* ESP32, C/C++, sensors, embedded

**Data Recovery & Photo Organizer** — Python tool for file recovery and automatic organization after data loss — grouping photos by EXIF metadata, duplicate detection, and date-based folder structure.

*Tech:* Python, PIL/Pillow, EXIF metadata, PhotoRec, file system automation

**movie-planner** — Cloud-based watchlist app with real-time sync — personal movie list featuring a modern GUI.

*Tech:* Python, Firebase (cloud DB), CustomTkinter

All projects: [github.com/karagiancode](https://github.com/karagiancode)

## EDUCATION

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**International Hellenic University (IHU)** — Thessaloniki

2019 – Present

*Department of Information & Electronic Systems Engineering — Specialization: Computer Science & Programming*

**30th General Lyceum of Thessaloniki**

2016 – 2019

*High School Diploma*

## CERTIFICATIONS, LANGUAGES & MORE

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- Certifications: International Diploma in IT Skills (Cambridge) · Basic Life Support / BLS (ERC)
- Languages: Greek (native) · English (B2 — good command)
- Volunteering: Upcoming volunteer work with the TechSaloniki organization (2026)
- Soft skills: Leadership, teamwork, working under pressure, communication, eager to learn and adapt
- Interests: Tae Kwon Do (WTF, 2013 – 2018), Chess, Gymnastics